

SIXTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

STATISTICAL QUALITY ASSURANCE IN TEXTILES

[Time: 3 hours]

(Maximum marks: 100)

*(Graph paper should be supplied)***PART-A**

(Maximum marks: 10)

Marks

I Answer the following questions in one or two sentences:

2 Marks each

1. State the term Mean?
2. Define the term sample?
3. What is standard error of SD?
4. What is a quality control chart?
5. What is mean by snap survey?

(5 x 2 = 10)

PART-B

(Maximum marks-30)

II (Answer any five of the following question. Each question carries 6 marks)

1. State the terms MD, SD and CV%?
2. Mention continuous frequency distribution with an example?
3. What are the features of a bell shaped distribution curve?
4. What is significance test and the Probability theory?
5. Describe advantages & uses of Quality control charts?
6. Write short note on:-
 - a. Warning limit
 - b. Action limit
7. Mention the Snap survey techniques

PART-C

(Answer one full question from each module. Each question carries 15 marks)

MODULE-I

III (a) State the terms range, Mean range & PMR?

(3)

(b) Draw Histogram, Frequency Polygon and different frequency curves with sketches (12)
with the help of an example?

OR

IV (a) Find out the Mean, SD & CV% from the frequency distribution of a count test value
given below? (15)

Class interval	Frequency
22 - 23	1
24 - 25	4
26 - 27	6
28 - 29	11
30 - 31	18
32 - 33	22
34 - 35	16
36 - 37	12
38 - 39	9
40 - 41	1
Total	100

MODULE-II

V 16 ring cops are tested for count and the mean found to be 44.1s. The S D of the 16 result (15)
is 2.2s.If the nominal count is 40s ,is the frame spinning too fine ?

OR

VI A filament yarn was tested for strength before and after chemical treatment. In each case (15)
8 specimens were tested and the following results were obtained:-

	Untreated filament	Treated filament
No. Of tests	8	8
Mean strength (gms.))	221	198
S D	6	10

Is there sufficient evidence to show that the chemical treatment has already
weakened the filament

MODULE- III

VII (a) Mention the specification limits in control charts? (3)

(b) Explain the steps for the preparation of a Mean quality control chart and Range Chart with equations? (12)

OR

VIII. VII Draw the Mean & Range charts from the given data in the table collected from a mill and state your comments ? (15)

Sample No.	1	2	3	4	5	6	7	8	9	10
Sample Mean(gms.)	11.2	12.7	12.4	12.3	12.2	11.7	12.1	11.9	12.4	12.3
Sample Range(gms)	1.8	2.3	2.5	2.2	1.8	1.9	2.1	2.4	1.8	2.1

(Given are $n = 4$, $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$)

MODULE-IV

IX (a) Mention the term Total Quality control? (3)

(b) Explain the. Maintenance of SQC systems adopted in Textile mills? (12)

OR

X (a) State the terms Cost reduction and quality control? (3)

(b) Describe the Specification of quality standards in a spinning Mills? (12)